Boris Andrews CV

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Employment

2025 – 2027 (*predicted*) **Postdoctoral Research Associate**, *Numerical Analysis*, University of Oxford
- Project: *ERC Starting Grant for Geometric Finite Element Methods (GeoFEM)*
- Advisor: *Kaibo Hu*

Education

2021 – 2025 **PhD (DPhil)**, *Numerical Analysis*, University of Oxford
- Thesis: *Geometric numerical integration via auxiliary variables*
- Supervisors: *Patrick Farrell*, *Wayne Arter*

2017 – 2021 **Integrated Masters (MMath)**, *Mathematics*, University of Oxford
- Grade: *First (Distinction)*
- Thesis: *Computation and approximation of near orthogonal matrices for tall random matrices*
- Supervisor: *Yuji Nakatsukasa*

Research Interests

Structure-preserving/compatible numerical methods for PDEs & ODEs, *Conservation & dissipation structures* | *Global & local energy estimates & conservation laws* | *Asymptotic-preserving integrators* | *Geometric machine learning*

Finite element theory, *Finite element exterior calculus (FEEC)* | *Domain decomposition* | *Parallel in time (PinT)*

Plasma modelling, *Magnetohydrodynamics (MHD)* | *Hybrid fluid-particle models*

Turbulent systems, *Stabilisation* | *Preconditioning*

Teaching

2026 – 2027 **Lecturer**, *Mixed Finite Element Methods and Finite Element Complexes*, Taught Course Centre (Bath, Bristol, Imperial, Oxford, Warwick, Swansea)
- 16-hour graduate course

2024 – 2025 **Tutor**, *Computational Mathematics*

2023 – 2024 **Tutor**, *Prelims corner*

Teaching assistant, *Numerical Linear Algebra*

2021 – 2022 **Teaching assistant**, *Random Matrix Theory*

Tutor, *Analysis I*, Oriel College

Supervision

Feb – Aug 2026 **Oscar Benton**, *MMSC dissertation*
- Project: *Numerical evolution of magnetic field topologies*

Feb – Aug 2025 **Matin Shams**, *MMSC special topic & dissertation*
- Project: *Enstrophy-stable integrators for the incompressible Navier–Stokes equations*

Sep – Oct 2024 **Sebastian Ohlig**, *Undergraduate summer internship*
- Project: *Stability study of conservative vs. symplectic integrators on the Toda lattice*

Publications & Preprints

Papers

Apr 2026 **Helicity-preserving finite element discretization for magnetic relaxation**, *with Mingdong He, Patrick Farrell, Kaibo Hu*
- Publication: *SIAM Journal on Scientific Computing (SISC), 48 (2), pp. B165-B183*

Dec 2025 **Enforcing conservation laws and dissipation inequalities numerically via auxiliary variables**, *with Patrick Farrell*
- Publication: *SIAM Journal on Scientific Computing (SISC), 47 (6), pp. A3516–A3535*

Preprints

(In review)

Jun 2026 **Automated Galerkin time stepping in Irksome**, *with Pablo Brubeck, Patrick Farrell, Robert Kirby, Scott MacLachlan*
- In review: *SIAM Journal on Scientific Computing (SISC)*

May 2026 **Arbitrary-order structure-preserving discretizations for geometric curvature flows**, *with Ganghui Zhang, Patrick Farrell*
- In review: *SIAM Journal on Scientific Computing (SISC)*

Nov 2025 **Conservative and dissipative discretisations of multi-conservative ODEs and GENERIC systems**, *with Patrick Farrell*
- In review: *Computers & Mathematics with Applications (CAMWA)*

Other works

Jul 2025 **Geometric numerical integration via auxiliary variables**
- Note: *PhD (DPhil) thesis*

TALKS (*upcoming)

INVITED TALKS & MINISYMPOSIUM PRESENTATIONS

2027 **ICOSAHOM*** *(Politecnico di Milano)* | **Oberwolfach Workshop on Structure-Preserving Methods*** *(Oberwolfach Research Institute for Mathematics)*

2026 **SciCADE*** *(University of Edinburgh)* | **ECCOMAS World Congress on Computational Mechanics*** *(Munich, Germany)* | **Invited Seminar** *(Chinese Academy of Sciences)*

2025 **(2×)ACOMEN** *(Ghent University)* | **ECCOMAS Thematic Conference on Modern Finite Element Technologies** *(Aachen, Germany)* | **Self-Consistency Group Seminar** *(CHaRMNET)* | **ACM Colloquium** *(University of Edinburgh × Heriot-Watt University)* | **Numerical Mathematics & Scientific Computing Seminar** *(Rice University)* | **SIAM CSE** *(Fort Worth, Texas)* | **Scientific Computing Seminar** *(Brown University)* | **METHODS Group Seminar** *(Brown University)*

2024 **External Seminar** *(Rice University)*

OTHER SEMINAR, WORKSHOP & CONFERENCE PRESENTATIONS

2026 **Firedrake User Meeting*** *(University of Oxford)* | **Programme on Differential Complexes** *(University of Vienna)* | **Workshop on Finite Element Tensor Calculus** *(Tsinghua University)*

2025 **Numerical Analysis Group Internal Seminar** *(University of Oxford)* | **Biennial Numerical Analysis Conference** *(University of Strathclyde)*

2024 **Computing Division Technical Meeting** *(UKAEA)* | **Firedrake User Meeting** *(University of Oxford)* | **PDEsoft** *(University of Cambridge)* | **European Finite Element Fair** *(University College London)* | **Exploiting Algebraic and Geometric Structure in Time-integration Methods Workshop** *(University of Pisa)* | **UKAEA PhD Student Engagement Day** *(UKAEA)* | **Junior Applied Mathematical Seminar** *(University of Warwick)*

2023 **ICIAM** *(Waseda University)* | **Numerical Analysis Group Internal Seminar** *(University of Oxford)* | **Junior Applied Mathematics Seminar** *(University of Oxford)* | **Met Office Presentation** *(University of Oxford)*

2022 **PRISM Workshop** *(Missenden Abbey, UK)*

Hosted Minisymposia

Jul 2026 **SciCADE**, *University of Edinburgh*
- Topic: *Topology-preserving finite element methods for magnetohydrodynamics*
- Co-organisers: *Patrick Farrell*, *Kaibo Hu*
- Speakers: *Tobias Blickhan*, *Mingdong He*, *Gunnar Hornig*, *Shipeng Mao*, *Adelle Wright*, *Anthony Yeates*, *Qian Zhang*

Jun 2025 **Biennial Numerical Analysis Conference**, *University of Strathclyde*
- Topic: *Structure-preserving finite element methods*
- Co-organiser: *Charlie Parker*
- Speakers: *Aaron Brunk*, *Mingdong He*, *Kaibo Hu*, *Rami Masri*

Other Experience

Sep – Dec 2026 **ETH Zurich**, *Co-editor of the Nachdiplom Lectures on Finite Element Tensor Calculus with Kaibo Hu*
- Contributed volume in preparation for publication

2025 – present **University of Oxford**, *Organisation of the Numerical Analysis Group's weekly finite element methods reading group*

Apr – Jun 2026 **University of Vienna**, *Attendance at the Programme on Differential Complexes at the Erwin Schrödinger International Institute for Mathematics and Physics (ESI)*

Aug – Oct 2022 **Tokamak Energy**, *Internship*, Physics: theory and modelling
- Project: *Implementation of non-Maxwellian backgrounds in the GENE gyrokinetic code*
- Supervisor: *Salomon Janhunen*

Jul 2022 **United Kingdom Atomic Energy Authority (UKAEA)**, *Plasma physics summer school*

Languages

Programming

Experienced: Python *(Firedrake)*, MATLAB, LaTeX | **Limited:** Julia, C, Fortran

Spoken

Fluent: English | **Intermediate:** Dutch | **Beginner:** Japanese, German